

EDUCATION

Master of Science (MS), Computer Science, Artificial Intelligence track *Aug'24 - May'26 (Expected)*
Binghamton University, State University of New York **GPA: 3.57/4.0**

Bachelor of Technology (B.Tech), Electronics & Communication Engineering *Jul'15 - June'19*
University of Engineering & Management, Kolkata, India **GPA: 3.20/4.0**

SKILLS

Programming Languages	Python, Java, C++, C, SQL, JavaScript, Golang, Kotlin, Typescript, React, R, HTML
Frameworks & Tools	PyTorch, ROS 2, Gazebo, .NET Framework, ServiceNow, PyCharm, Node.js
Cloud & DevOps	AWS (EC2, S3), Azure, GCP, Docker, Kubernetes
Databases	MySQL, SQL Server Reporting Services

EXPERIENCE

Software Engineer Intern, Global Health Impact *May'25 - Aug'25*

- Automated ingestion of program data from Google Sheets and built a scalable entity-mapping pipeline using Python and Pandas, improving data integration accuracy and cutting manual preprocessing time by 75% [**Pandas**]
- Designed and implemented new frontend components for the organization's website, including interactive dashboards and responsive layouts, enhancing usability and engagement for global stakeholders [JavaScript, HTML, CSS]

Graduate Research Assistant, Binghamton University *Aug'24 - May'25*

- Designed and evaluated fair-division allocation algorithms (Randomized, Round-Robin, Max-Social-Welfare) to study agent satisfaction under EF1 constraints using LLM-simulated agents.
- Built a scalable Python + Redis system handling 10,000+ LLM calls/day, validating 3,000+ responses, automating 1,000+ allocations, reducing runtime to 0.4s, and exposing fairness metrics through Tableau dashboards.
- Deployed the system for real CS graduate course registration, assigning students to oversubscribed courses based on preferences.

Software Engineer, Accenture Solutions Pvt. Ltd *Jan'20 - Sep'22*

- Automated customer service intake workflows by developing custom ServiceNow applications using JavaScript and platform APIs, increasing request throughput by 15% and reducing turnaround time by 20%.
- Built SSRS dashboards backed by .NET services to generate performance and operational reports, cutting report generation time by 30% (10 → 7 minutes per report).
- Designed and deployed a Python-based incident classification system that processed 5,000+ incidents/month, automatically categorizing by priority, issue type, and region, saving 4,000 engineering hours annually.
- Led end-to-end migration of 120 on-prem servers in Kenya to AWS (EC2, S3) using infrastructure-as-code templates, eliminating power-failure downtime and reducing operating costs by \$190K/year.
- Automated decommissioning of 300 legacy servers using Python scripts, reducing manual effort by 50% and cutting operational expenses by \$50K annually.
- Upgraded and managed 200+ Windows servers using PowerShell DSC and Active Directory integration, improving security and reliability while maintaining 99.9% uptime.

PROJECTS

Intelligent Robotic Conversational & Interaction System 🤖 *Jan'25 - May'25*

- Built a ROS 2-based museum tour guide using the Tiago robot in Gazebo, achieving 97% navigation success across 40+ dynamic goals with sub-second replanning.
- Integrated GPT-4 for multi-turn, context-aware dialogue across concurrent users, delivering low-latency (1.8s) responses and high intent accuracy (95%).
- Implemented YOLOv8 anomaly detection and speech recognition (Google STT) to enable real-time crowd-aware rerouting and personalized tour adjustments.

Cross-Platform Meme Analysis System 🗣️ *Aug'25 - Dec'25*

- Built a FastAPI backend and multi-page Streamlit dashboard to analyze meme diffusion, semantic drift, and cross-platform influence between 4chan and Reddit in real time.
- Integrated OpenAI ChatGPT API for LLM-assisted explanations, making complex analytics interpretable for end users.
- Enhanced usability with persistent session state, custom theming, and interactive Plotly visualizations, delivering a polished, production-ready research tool.

ACHIEVEMENTS

- Accenture Celebrates Excellence Award (Q3 2021)**: Recognized for cloud migration project that saved client \$500K annually by transitioning on-prem servers to Azure/AWS
- Performance Achievement Award (2022)**: Orchestrated VCenter upgrade and developed a mentoring program for 50+ new hires, reducing onboarding time by 2 weeks